

**WQD 7009**  
**BIG DATA APPLICATION AND ANALYTICS**

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**ASSIGNMENT 1**

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## PART A: INTRODUCTION

The dataset chosen is the annual greenhouse gas emissions by activity and by region from IMS Climate Change Dashboard [\[1\]](#). Regional annual Air Emission Accounts are presented by industry and household for four gases - Carbon Dioxide, methane, nitrous oxide, and F-gases. The F-gases constitute of hydrofluorocarbons, perfluorocarbons, sulfur hexafluoride and nitrogen trifluoride. Emissions are presented in Million metric tons of CO<sub>2</sub> equivalent (MTCARBON DIOXIDEe).

This dataset consists of 1186 records containing detailed data on greenhouse gas emissions across various countries, industries, and gas types over multiple years. The dataset includes 27 attributes that provide a comprehensive view of emissions, sources, and classifications. The data is designed to support analysis of global emissions trends, industry contributions, and country-level comparisons. The key attributes included in the dataset are:

| NO | ATTRIBUTES | DESCRIPTION                                             | UNIT/RECORD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DATA TYPE |
|----|------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1  | Objectld2  | A unique identifier for each record.                    | General Numbers from 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Integer   |
| 2  | Country    | Emissions data is provided for 25 countries or regions. | 1. Advanced Economies<br>2. Africa<br>3. Americas<br>4. Asia<br>5. Australia and New Zealand<br>6. Central Asia<br>7. Eastern Asia<br>8. Eastern Europe<br>9. Emerging and Developing Economies<br>10. Europe<br>11. G20<br>12. G7<br>13. Latin America and the Caribbean<br>14. Northern Africa<br>15. Northern America<br>16. Northern Europe<br>17. Oceania<br>18. Other Oceania sub-regions<br>19. South-eastern Asia<br>20. Southern Asia<br>21. Southern Europe<br>22. Sub-Saharan Africa<br>23. Western Asia<br>24. Western Europe<br>25. World | String    |
| 3  | ISO2       | No values displayed                                     | Blanks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Null      |
| 4  | ISO3       | Codes to identify countries                             | 1. AETMP<br>2. NA605<br>3. AMETMP<br>4. ASIATMP<br>5. ASIATMP<br>6. NACA                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | String    |

| NO | ATTRIBUTES          | DESCRIPTION                                                   | UNIT/RECORD                                                                                                                                                                                                                                                                                                                                               | DATA TYPE |
|----|---------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|    |                     |                                                               | 7. NA510<br>9. EMDETMP<br>11. NA120<br>13. LACTMP<br>15. NA225<br>17. OCETMP<br>19. NASEA<br>21. NASE<br>23. NAWA<br>26. WLD<br>8. NAEE<br>10. EURTMP<br>12. NA119<br>14. NANA9<br>16. NANE<br>18. NAOOSR<br>20. NASA<br>22. SSA<br>24. NAWE                                                                                                              |           |
| 5  | Indicator           | The emissions indicator being measured                        | Annual greenhouse gas (GHG) air emissions accounts                                                                                                                                                                                                                                                                                                        | String    |
| 6  | Unit                | The unit of measurement for emissions                         | Million metric tons of CARBON DIOXIDE equivalent                                                                                                                                                                                                                                                                                                          | String    |
| 7  | Source              | The source of the emissions data                              | Organisation for Economic Co-operation and Development                                                                                                                                                                                                                                                                                                    | String    |
| 8  | CTS Code            | The code representing the specific sector or emission source. | ECNGA                                                                                                                                                                                                                                                                                                                                                     | String    |
| 9  | CTS Name            | The name of the sector or emission source.                    | Greenhouse Gas Emissions (GHG); Air Emissions Accounts                                                                                                                                                                                                                                                                                                    | String    |
| 10 | CTS Full Descriptor | A detailed description of the sector or emission source.      | Environment, Climate Change, Greenhouse Gas Emissions, Air Emissions Accounts                                                                                                                                                                                                                                                                             | String    |
| 11 | Industry            | The industry sector related to the emissions                  | 1. Agriculture, Forestry and Fishing<br>2. Construction<br>3. Electricity, Gas, Steam and Air Conditioning Supply<br>4. Manufacturing<br>5. Mining<br>6. Other Services Industries<br>7. Total Households<br>8. Total Industry and Households<br>9. Transportation and Storage<br>10. Water supply; sewerage, waste management and remediation activities | String    |
| 12 | Gas Type            | The type of greenhouse gas                                    | 1. Carbon Dioxide                                                                                                                                                                                                                                                                                                                                         | String    |

| NO | ATTRIBUTES              | DESCRIPTION                                                                | UNIT/RECORD                                                                 | DATA TYPE |
|----|-------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------|
|    |                         |                                                                            | 2. Fluorinated gases<br>3. Greenhouse gas<br>4. Methane<br>5. Nitrous oxide |           |
| 13 | Seasonal Adjustment     | Information on whether the data has been seasonally adjusted.              | Seasonally Adjusted                                                         | String    |
| 14 | Scale                   | The scale of the emissions data                                            | Units                                                                       | String    |
| 15 | Record for 2010 to 2023 | Emissions data from 2010 to 2023 for each gas type, country, and industry. | Numerical (Values)                                                          | Float     |

Table 1: Description of All The Key Attributes In The Original Dataset

For this analysis, the focus is placed on 5 key attributes: Object ID, Country, Gas Type, Industry, and emissions data for the years 2022 and 2023. These attributes are used to derive meaningful insights regarding recent trends in greenhouse gas emissions, with an emphasis on understanding the contributions of specific industries and countries to the global emissions landscape.

| NO | ATTRIBUTE | DESCRIPTION                                                                     | ORIGINAL NAME                                                                                               | AMENDED NAME                                               |
|----|-----------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1. | Obejct ID | The unique identifier for each record, allowing us to track individual entries. | General numbers from 1                                                                                      | N/A                                                        |
| 2. | Country   | Only 6 countries were used for this analysis.                                   | 1. Africa<br>2. Asia<br>3. Australia and New Zealand<br>4. Eastern Asia<br>5. Europe<br>6. Northern America | 1. Africa<br>2. Asia<br>3. ANZ<br>4. EA<br>5. EU<br>6. NAM |
| 3. | Gas Type  | Only 4 gas type were used for this analysis.                                    | 1. Carbon Dioxide<br>2. Fluorinated gases<br>3. Methane<br>4. Nitrous oxide                                 | 1. CO2<br>2. FG<br>3. CH4<br>4. N2O                        |
| 4. | Industry  | To categorize the emissions data by industry sector                             | Agriculture, Forestry and Fishing                                                                           | Agriculture                                                |

| NO | ATTRIBUTE | DESCRIPTION                                                                                      | ORIGINAL NAME                                                       | AMENDED NAME           |
|----|-----------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------|
|    |           |                                                                                                  | Water supply; sewerage, waste management and remediation activities | WWS                    |
|    |           |                                                                                                  | Construction                                                        | Construction           |
|    |           |                                                                                                  | Electricity, Gas, Steam and Air Conditioning Supply                 | EnergyCS               |
|    |           |                                                                                                  | Manufacturing                                                       | Manuf                  |
|    |           |                                                                                                  | Mining                                                              | Mining                 |
|    |           |                                                                                                  | Other Services Industries                                           | Others                 |
|    |           |                                                                                                  | Total Households                                                    | Households             |
|    |           |                                                                                                  | Total Industry and Households                                       | Ind_HH                 |
|    |           |                                                                                                  | Transportation and Storage                                          | Trans_Str              |
| 5. | Year      | The emissions data for the most recent years, which provide a snapshot of current global trends. | 1. 2022<br>2. 2023                                                  | 1. Y_2022<br>2. Y_2023 |

Table 2: Description for all the 5 Attributes chosen.

After eliminating irrelevant columns and records, the focus has been narrowed to a smaller, more relevant subset that provides valuable insights into the most recent emissions data. Now the dataset is ready with 223 records and 5 key attributes as mentioned in Table 2.

## PART B: USAGE OF HIVE

The Hive table was first created as part of the data preparation workflow to organize and structure the greenhouse gas emissions data before importing it into HBase. The following commands were used to create a structured table in Hive as part of the initial data processing pipeline, before the data was imported into HBase.

a. Please refer to the screenshot below for the output.

```
hive> show databases;
OK
assignment
assignment1
default
task
test
Time taken: 0.018 seconds, Fetched: 5 row(s)
hive> use task;
OK
Time taken: 0.010 seconds
hive> CREATE TABLE GreenHouseGas (Object_ID INT, Country STRING, Industry STRING, Gas_Type STRING, Y_2022 INT, Y_2023 INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY "," STORED AS TEXTFILE;
OK Gas_Type STRING, Y_2022 INT, Y_2023 INT) ROW FORMAT DELIMITED FIELDS TERMINAT
Time taken: 0.157 secondsILE;
hive> CREATE TABLE GreenHouseGas (Object_ID INT, Country STRING, Industry STRING, Gas_Type STRING, Y_2022 INT, Y_2023 INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY "," STORED AS TEXTFILE;
FAILED: Execution Error, return code 1 from org.apache.hadoop.hive.ql.exec.DDLTask. AlreadyExistsException(message:Table GreenHouseGas already exists)
hive> show tables;
OK
greenhousegas
Time taken: 0.020 seconds, Fetched: 1 row(s)
hive> load data local inpath '/home/cloudera/Downloads/Task_1.csv' into table GreenHouseGas;
Loading data to table task.greenhousegas
Table task.greenhousegas stats: [numFiles=1, totalSize=6121]
OK
Time taken: 0.428 seconds
hive> select * from GreenHouseGas limit 10;
OK
1      Africa  Agriculture  CO2      19      18
2      Africa  Agriculture  CH4      669     656
3      Africa  Agriculture  N2O      264     265
4      Africa  Construction CO2      16      15
5      Africa  Construction CH4      0       0
6      Africa  Construction N2O      0       0
7      Africa  EnergyCS    CO2      464     428
8      Africa  EnergyCS    CH4      0       0
9      Africa  EnergyCS    N2O      10      10
10     Africa  Manuf      CO2      345     330
Time taken: 0.057 seconds, Fetched: 10 row(s)
```

b. Next, the table created in HIVE was imported into HBase using the following commands. To enable this process, the following four JAR files were added.

```
hive> add jar /usr/lib/hive/lib/zookeeper.jar;
Added [/usr/lib/hive/lib/zookeeper.jar] to class path
Added resources: [/usr/lib/hive/lib/zookeeper.jar]
hive> add jar /usr/lib/hive/lib/hbase-client.jar;
Added [/usr/lib/hive/lib/hbase-client.jar] to class path
Added resources: [/usr/lib/hive/lib/hbase-client.jar]
hive> add jar /usr/lib/hive/lib/guava-14.0.1.jar;
Added [/usr/lib/hive/lib/guava-14.0.1.jar] to class path
Added resources: [/usr/lib/hive/lib/guava-14.0.1.jar]
hive> add jar /usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.13.0.jar;
/usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.13.0.jar does not exist
Query returned non-zero code: 1, cause: /usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.13.0.jar does not exist
hive> add jar /usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.10.0.jar;
Added [/usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.10.0.jar] to class path
Added resources: [/usr/lib/hive/lib/hive-hbase-handler-1.1.0-cdh5.10.0.jar]
hive> CREATE TABLE GreenHouseGasAnnual (Object ID INT, Country STRING, Industry STRING, Gas Type STRING, Y_2022 INT, Y_2023 INT) STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler' WITH SERDEPROPERTIES ('hbase.columns.mapping' =
```

c. A mapping table named GreenHouseGasAnnual was created in both HIVE and HBase using the commands highlighted in the green box.

```
hive> CREATE TABLE GreenHouseGasAnnual (Object ID INT, Country STRING, Industry STRING, Gas Type STRING, Y_2022 INT, Y_2023 INT) STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler' WITH SERDEPROPERTIES ('hbase.columns.mapping' =
":key,cf:Country,cf:Industry,cf:Gas_Type,cf:Y_2022,cf:Y_2023") TBLPROPERTIES ('hbase.table.name' = 'GreenHouseGasAnnual');
OK
Time taken: 1.456 seconds
hive> insert into table GreenHouseGasAnnual select *from GreenHouseGas;
Query ID = cloudera_20241130035555_b720aee60-8b57-46cb-8718-eab53b373016
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1732879953013_0005, Tracking URL = http://quickstart.cloudera:8080/proxy/application_1732879953013_0005/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1732879953013_0005
```



- d. The data from the GreenHouseGas table was inserted into the GreenHouseGasAnnual table. To view the content in HIVE, the SELECT \* FROM query was used with a limit of 10 rows.

Time taken: 1.456 seconds

```
hive> insert into table GreenHouseGasAnnual select *from GreenHouseGas;
```

Query ID = cloudera\_20241130035555\_b269ae60-0b52-46cb-9218-eab53b323916

Total jobs = 1

Launching Job 1 out of 1

Number of reduce tasks is set to 0 since there's no reduce operator

Starting Job = job\_1732879953013\_0005, Tracking URL = http://quickstart.cloudera:8088/proxy/application\_1732879953013\_0005/

Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job\_1732879953013\_0005

Hadoop job information for Stage-0: number of mappers: 1; number of reducers: 0

2024-11-30 03:55:19,894 Stage-0 map = 0%, reduce = 0%

2024-11-30 03:55:34,240 Stage-0 map = 100%, reduce = 0%, Cumulative CPU 2.9 sec

MapReduce Total cumulative CPU time: 2 seconds 900 msec

Ended Job = job\_1732879953013\_0005

MapReduce Jobs Launched:

Stage-Stage-0: Map: 1 Cumulative CPU: 2.9 sec HDFS Read: 10221 HDFS Write: 0 SUCCESS

Total MapReduce CPU Time Spent: 2 seconds 900 msec

OK

Time taken: 29.456 seconds

```
hive> select * from GreenHouseGasAnnual limit 10;
```

OK

|     |        |             |     |     |     |
|-----|--------|-------------|-----|-----|-----|
| 1   | Africa | Agriculture | CO2 | 19  | 18  |
| 10  | Africa | Manuf       | CO2 | 345 | 330 |
| 100 | ANZ    | Households  | CH4 | 1   | 2   |
| 101 | ANZ    | Households  | N2O | 1   | 1   |
| 102 | ANZ    | Ind_HH      | CO2 | 443 | 449 |
| 103 | ANZ    | Ind_HH      | FG  | 15  | 15  |
| 104 | ANZ    | Ind_HH      | CH4 | 143 | 154 |
| 105 | ANZ    | Ind_HH      | N2O | 26  | 26  |
| 106 | ANZ    | Trans_Str   | CO2 | 80  | 82  |
| 107 | ANZ    | Trans_Str   | FG  | 2   | 2   |

Time taken: 0.138 seconds, Fetched: 10 row(s)

hive> █



- e. Checked Hue to verify if the file exists in Hbase. The file was confirmed to be present in the HBase environment.

Hue - HBase Browser

quickstart.cloudera:8888/hbase/#Cluster/GreenHouseGasAnnual

Cloudera Hue Hadoop HBase Impala Spark Solr Oozie Cloudera Manager Getting Started

HUE Home Query Editors Data Browsers Workflows Search Security

HBase Browser

Home - Cluster / GreenHouseGasAnnual

row\_key, row\_prefix\* +scan\_len [col1, family:col2, fam3:, col\_prefix\* +3, fam: col2 to col3] {Filter1() AND cf:}

| cf: Y_2022 | cf: Country | cf: Gas_Type | cf: Industry | cf: Y_2023 |  |
|------------|-------------|--------------|--------------|------------|--|
| 19         | Africa      | CO2          | Agriculture  | 18         |  |
| 345        | Africa      | CO2          | Manuf        | 330        |  |
| 1          | ANZ         | CH4          | Households   | 2          |  |
| 1          | ANZ         | N2O          | Households   | 1          |  |
|            |             |              |              |            |  |
|            |             |              |              |            |  |

10 entries starting from null in 3.491 seconds.

[cloudera@quickstart:~] [cloudera@quickstart:~] Hue - HBase Browser - ...

## PART C: 5 SHELL COMMANDS

### 1. LIST

- The list command was used to view the tables in HBase. The GreenHouseGasAnnual table was successfully created in HBase.

```
hbase(main):004:0> list
TABLE
GHG_EMISSION_ANL
GreenHouseGasAnnual
2 row(s) in 0.0110 seconds
=> ["GHG_EMISSION_ANL", "GreenHouseGasAnnual"]
hbase(main):005:0>
```



- The scan command was used to view the contents of the table.

```
=> ["GHG_EMISSION_ANL", "GreenHouseGasAnnual"]
hbase(main):005:0> scan 'GreenHouseGasAnnual'
```

- After pre-processing in Excel, the dataset contains 223 rows and was successfully imported into the HBase table GreenHouseGasAnnual.

```
99
223 row(s) in 1.4310 seconds
hbase(main):006:0>
```



- A portion of the dataset in HBase is provided as evidence.

```
222 column=cf:Y_2022, timestamp=1732967733211, value=145
222 column=cf:Y_2023, timestamp=1732967733211, value=150
223 column=cf:Country, timestamp=1732967733211, value=NAM
223 column=cf:Gas_Type, timestamp=1732967733211, value=M20
223 column=cf:Industry, timestamp=1732967733211, value=WW5
223 column=cf:Y_2022, timestamp=1732967733211, value=27
223 column=cf:Y_2023, timestamp=1732967733211, value=28
23 column=cf:Country, timestamp=1732967733211, value=Africa
23 column=cf:Gas_Type, timestamp=1732967733211, value=M20
23 column=cf:Industry, timestamp=1732967733211, value=Households
23 column=cf:Y_2022, timestamp=1732967733211, value=13
23 column=cf:Y_2023, timestamp=1732967733211, value=14
24 column=cf:Country, timestamp=1732967733211, value=Africa
24 column=cf:Gas_Type, timestamp=1732967733211, value=C02
24 column=cf:Industry, timestamp=1732967733211, value=Ind_HH
24 column=cf:Y_2022, timestamp=1732967733211, value=1485
24 column=cf:Y_2023, timestamp=1732967733211, value=1438
25 column=cf:Country, timestamp=1732967733211, value=Africa
25 column=cf:Gas_Type, timestamp=1732967733211, value=F6
25 column=cf:Industry, timestamp=1732967733211, value=Ind_HH
25 column=cf:Y_2022, timestamp=1732967733211, value=83
25 column=cf:Y_2023, timestamp=1732967733211, value=88
26 column=cf:Country, timestamp=1732967733211, value=Africa
26 column=cf:Gas_Type, timestamp=1732967733211, value=CH4
```

## 2. WHOAMI

This HBase command retrieves the current user's information, displaying 'cloudera' as the default user.

```
hbase(main):014:0> whoami
cloudera (auth:SIMPLE)
  groups: cloudera, default
hbase(main):015:0> █
```

## 3. STATUS

This command provides details about the HBase cluster, including its overall health, the number of servers, and regions. In my case, there is 1 active master and 1 server.

```
hbase(main):015:0> status
1 active master, 0 backup masters, 1 servers, 0 dead, 4.0000 average load
hbase(main):016:0> █
```

## 4. VERSION

This command retrieves the version information of HBase. The installed and currently used version of HBase is 1.2.0-cdh5.10.0.

```
hbase(main):016:0> version
1.2.0-cdh5.10.0, rUnknown, Fri Jan 20 12:13:18 PST 2017
hbase(main):017:0> █
```

## 5. TABLE\_HELP

The list of available commands related to table operations was viewed before executing the DML commands, to gain a better understanding.

```

hbase(main):017:0> table_help
Help for table-reference commands.

You can either create a table via 'create' and then manipulate the table via commands like 'put', 'get', etc.
See the standard help information for how to use each of these commands.

However, as of 0.96, you can also get a reference to a table, on which you can invoke commands.
For instance, you can get create a table and keep around a reference to it via:

hbase> t = create 't', 'cf'

Or, if you have already created the table, you can get a reference to it:

hbase> t = get_table 't'

You can do things like call 'put' on the table:

hbase> t.put 'r', 'cf:q', 'v'

which puts a row 'r' with column family 'cf', qualifier 'q' and value 'v' into table t.

To read the data out, you can scan the table:

hbase> t.scan

which will read all the rows in table 't'.

Essentially, any command that takes a table name can also be done via table reference.
Other commands include things like: get, delete, deleteall,
get all columns, get counter, count, incr. These functions, along with
the standard JRuby object methods are also available via tab completion.

For more information on how to use each of these commands, you can also just type:

hbase> t.help 'scan'

which will output more information on how to use that command.

You can also do general admin actions directly on a table; things like enable, disable,
flush and drop just by typing:

hbase> t.enable
hbase> t.flush
hbase> t.disable

```



## 6. COUNT

This command is used to count the number of rows, indicating 223 records.

```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
ed. Instead, use io.native.lib.available  
HBase Shell; enter 'help<RETURN>' for list of supported commands.  
Type "exit<RETURN>" to leave the HBase Shell  
Version 1.2.0-cdh5.10.0, rUnknown, Fri Jan 20 12:13:18 PST 2017  
  
hbase(main):001:0> list  
TABLE  
0 row(s) in 0.3120 seconds  
  
=> []  
hbase(main):002:0> list  
TABLE  
GreenHouseGasAnnual  
1 row(s) in 0.0200 seconds  
  
=> ["GreenHouseGasAnnual"]  
hbase(main):003:0> count rows  
NameError: undefined local variable or method `rows' for #<Object:0x1bd8dc2e>  
  
hbase(main):004:0> count 'GreenHouseGasAnnual'  
223 row(s) in 0.5690 seconds  
  
=> 223  
hbase(main):005:0> █
```

## PART D: HBASE COMMANDS

### 5 Data Definition Language and 10 Data Manipulation Language Commands

#### 1. DESCRIBE (DDL)

This command is used to describe the table created in HBase. Notably, the column family created is 'cf', and it is displayed accordingly.

```
hbase(main):013:0> describe 'GreenHouseGasAnnual'
Table GreenHouseGasAnnual is ENABLED
GreenHouseGasAnnual
COLUMN FAMILIES DESCRIPTION
{NAME => 'cf', DATA_BLOCK_ENCODING => 'NONE', BLOOMFILTER => 'ROW', REPLICATION
SCOPE => '0', VERSIONS => '1', COMPRESSION => 'NONE', MIN_VERSIONS => '0', TTL =
> 'FOREVER', KEEP_DELETED_CELLS => 'FALSE', BLOCKSIZE => '65536', IN_MEMORY => '
false', BLOCKCACHE => 'true'}
1 row(s) in 0.0230 seconds

hbase(main):014:0>
```

#### 2. GET (DML)

The get command was used to verify that the data records are in order and reflect the records from the CSV file, confirming their accuracy.

```
hbase(main):007:0> get 'GreenHouseGasAnnual', '223'
COLUMN                                CELL
cf:Country                            timestamp=1732967733211, value=NAM
cf:Gas_Type                           timestamp=1732967733211, value=N2O
cf:Industry                           timestamp=1732967733211, value=WNS
cf:Y_2022                             timestamp=1732967733211, value=27
cf:Y_2023                             timestamp=1732967733211, value=28
5 row(s) in 0.0040 seconds

hbase(main):008:0> get 'GreenHouseGasAnnual', '1'
COLUMN                                CELL
cf:Country                            timestamp=1732967733211, value=Africa
cf:Gas_Type                           timestamp=1732967733211, value=CO2
cf:Industry                           timestamp=1732967733211, value=Agriculture
cf:Y_2022                             timestamp=1732967733211, value=19
cf:Y_2023                             timestamp=1732967733211, value=18
5 row(s) in 0.0060 seconds

hbase(main):009:0> get 'GreenHouseGasAnnual', '99'
COLUMN                                CELL
cf:Country                            timestamp=1732967733211, value=ANZ
cf:Gas_Type                           timestamp=1732967733211, value=FG
cf:Industry                           timestamp=1732967733211, value=Households
cf:Y_2022                             timestamp=1732967733211, value=2
cf:Y_2023                             timestamp=1732967733211, value=2
5 row(s) in 0.0040 seconds
```

Dataset in CSV file.

|     | A   | B      | C           | D   | E   | F   | G | H |
|-----|-----|--------|-------------|-----|-----|-----|---|---|
| 1   | 1   | Africa | Agriculture | CO2 | 19  | 18  |   |   |
| 2   | 2   | Africa | Agriculture | CH4 | 669 | 656 |   |   |
| 3   | 3   | Africa | Agriculture | N2O | 264 | 265 |   |   |
| 98  | 98  | ANZ    | Household   | CO2 | 61  | 62  |   |   |
| 99  | 99  | ANZ    | Household   | FG  | 2   | 2   |   |   |
| 100 | 100 | ANZ    | Household   | CH4 | 1   | 2   |   |   |
| 221 | 221 | NAM    | WWS         | CO2 | 27  | 27  |   |   |
| 222 | 222 | NAM    | WWS         | CH4 | 145 | 150 |   |   |
| 223 | 223 | NAM    | WWS         | N2O | 27  | 28  |   |   |

### 3. PUT (DML)

The put command was used to insert the total gas emissions for both 2022 and 2023 for all countries, gas types, and industries. Hence the total record is 224.

```
hbase(main):106:0> put 'GreenHouseGasAnnual', '224', 'cf:Country', 'All_Country/Contingent'
0 row(s) in 0.0940 seconds

hbase(main):107:0> put 'GreenHouseGasAnnual', '224', 'cf:Gas_Type', 'All_Gas_Type'
0 row(s) in 0.0190 seconds

hbase(main):108:0> put 'GreenHouseGasAnnual', '224', 'cf:Industry', 'All_Indsutry'
0 row(s) in 0.0510 seconds

hbase(main):109:0> put 'GreenHouseGasAnnual', '224', 'cf:Y_2022', '65,790'
0 row(s) in 0.0330 seconds

hbase(main):110:0> put 'GreenHouseGasAnnual', '224', 'cf:Y_2023', '65,577'
0 row(s) in 0.0770 seconds

hbase(main):111:0> get 'GreenHouseGasAnnual', '224'
COLUMN                                CELL
cf:Country                             timestamp=1732987871994, value=All_Country/Contingent
cf:Gas_Type                             timestamp=1732987878833, value=All_Gas_Type
cf:Industry                             timestamp=1732987884103, value=All_Indsutry
cf:Y_2022                               timestamp=1732987891459, value=65,790
cf:Y_2023                               timestamp=1732987898129, value=65,577
5 row(s) in 0.0150 seconds
```



#### 4. APPEND (DML)

The GHG indicator was added, as the All\_Gas\_Type represents the summation of all records (4 Types Gas).

```
hbase(main):112:0> append 'GreenHouseGasAnnual', '224', 'cf:Gas_Type', '(GHG)'
0 row(s) in 0.0940 seconds

hbase(main):113:0> get 'GreenHouseGasAnnual', '224'
COLUMN                                CELL
cf:Country                            timestamp=1732987871994, value=All_Country/Contingent
cf:Gas_Type                           timestamp=1732987927085, value=All_Gas_Type(GHG)
cf:Industry                           timestamp=1732987884103, value=All_Indsutry
cf:Y_2022                             timestamp=1732987891459, value=65,790
cf:Y_2023                             timestamp=1732987898129, value=65,577
5 row(s) in 0.0140 seconds
```

#### 5. SCAN (DML)

A series of SCAN command were used for analysis, based on the analysis done using Excel it reveals that Carbon Dioxide is the largest emitted gas in both 2022 and 2023. Therefore, the analysis is primarily focused on Carbon Dioxide emissions.

- a. The scan command/query was used to view the Carbon Dioxide records for Asia, the largest Carbon Dioxide producing region, and it displayed 10 records that match the data in my CSV file.

```
hbase(main):029:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Gas_Type',=, 'binary:C02')
) AND (SingleColumnValueFilter('cf','Country',=, 'binary:Asia'))"}

```

| ROW | COLUMN+CELL                                                     |
|-----|-----------------------------------------------------------------|
| 34  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 34  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 34  | column=cf:Industry, timestamp=1732967733211, value=Agriculture  |
| 34  | column=cf:Y_2022, timestamp=1732967733211, value=236            |
| 34  | column=cf:Y_2023, timestamp=1732967733211, value=238            |
| 38  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 38  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 38  | column=cf:Industry, timestamp=1732967733211, value=Construction |
| 38  | column=cf:Y_2022, timestamp=1732967733211, value=428            |
| 38  | column=cf:Y_2023, timestamp=1732967733211, value=439            |
| 42  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 42  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 42  | column=cf:Industry, timestamp=1732967733211, value=EnergyCS     |
| 42  | column=cf:Y_2022, timestamp=1732967733211, value=9769           |
| 42  | column=cf:Y_2023, timestamp=1732967733211, value=9785           |
| 46  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 46  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 46  | column=cf:Industry, timestamp=1732967733211, value=Manuf        |
| 46  | column=cf:Y_2022, timestamp=1732967733211, value=6823           |
| 46  | column=cf:Y_2023, timestamp=1732967733211, value=6827           |
| 50  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 50  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 50  | column=cf:Industry, timestamp=1732967733211, value=Mining       |
| 50  | column=cf:Y_2022, timestamp=1732967733211, value=1170           |
| 50  | column=cf:Y_2023, timestamp=1732967733211, value=1192           |
| 54  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 54  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 54  | column=cf:Industry, timestamp=1732967733211, value=Others       |
| 54  | column=cf:Y_2022, timestamp=1732967733211, value=402            |
| 54  | column=cf:Y_2023, timestamp=1732967733211, value=422            |
| 58  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 58  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 58  | column=cf:Industry, timestamp=1732967733211, value=Households   |
| 58  | column=cf:Y_2022, timestamp=1732967733211, value=1860           |
| 58  | column=cf:Y_2023, timestamp=1732967733211, value=1833           |
| 62  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 62  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 62  | column=cf:Industry, timestamp=1732967733211, value=Ind_HH       |
| 62  | column=cf:Y_2022, timestamp=1732967733211, value=22588          |
| 62  | column=cf:Y_2023, timestamp=1732967733211, value=22646          |
| 66  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 66  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |
| 66  | column=cf:Industry, timestamp=1732967733211, value=Trans_Str    |
| 66  | column=cf:Y_2022, timestamp=1732967733211, value=1813           |
| 66  | column=cf:Y_2023, timestamp=1732967733211, value=1820           |
| 70  | column=cf:Country, timestamp=1732967733211, value=Asia          |
| 70  | column=cf:Gas_Type, timestamp=1732967733211, value=C02          |

|    | A  | B       | C             | D   | E     | F     | G |
|----|----|---------|---------------|-----|-------|-------|---|
| 1  | ID | Country | Industry      | Gas | 20    | 20    |   |
| 35 | 34 | Asia    | Agriculture   | CO2 | 236   | 238   |   |
| 39 | 38 | Asia    | Construction  | CO2 | 428   | 439   |   |
| 43 | 42 | Asia    | EnergyCS      | CO2 | 9769  | 9785  |   |
| 47 | 46 | Asia    | Manufacturing | CO2 | 6823  | 6827  |   |
| 51 | 50 | Asia    | Mining        | CO2 | 1170  | 1192  |   |
| 55 | 54 | Asia    | Others        | CO2 | 402   | 422   |   |
| 59 | 58 | Asia    | Household     | CO2 | 1860  | 1833  |   |
| 63 | 62 | Asia    | Ind_HH        | CO2 | 22588 | 22646 |   |
| 67 | 66 | Asia    | Trans_Str     | CO2 | 1813  | 1820  |   |
| 71 | 70 | Asia    | WWS           | CO2 | 86    | 91    |   |
| 25 |    |         |               |     |       |       |   |

- b. The scan query was used to view all the records for the Electricity, Gas, Steam and Air Conditioning Supply (EnergyCS) industry in Asia, as it contributes the largest amount of greenhouse gas in the Asia region, with the records reflecting this.

```
hbase(main):005:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Country', =, 'binary:Asia')
) AND (SingleColumnValueFilter('cf', 'Industry', =, 'binary:EnergyCS'))"}
ROW COLUMN+CELL
42 column=cf:Country, timestamp=1732967733211, value=Asia
42 column=cf:Gas_Type, timestamp=1732967733211, value=CO2
42 column=cf:Industry, timestamp=1732967733211, value=EnergyCS
42 column=cf:Y_2022, timestamp=1732967733211, value=9769
42 column=cf:Y_2023, timestamp=1732967733211, value=9785
43 column=cf:Country, timestamp=1732967733211, value=Asia
43 column=cf:Gas_Type, timestamp=1732967733211, value=FG
43 column=cf:Industry, timestamp=1732967733211, value=EnergyCS
43 column=cf:Y_2022, timestamp=1732967733211, value=0
43 column=cf:Y_2023, timestamp=1732967733211, value=0
44 column=cf:Country, timestamp=1732967733211, value=Asia
44 column=cf:Gas_Type, timestamp=1732967733211, value=CH4
44 column=cf:Industry, timestamp=1732967733211, value=EnergyCS
44 column=cf:Y_2022, timestamp=1732967733211, value=17
44 column=cf:Y_2023, timestamp=1732967733211, value=18
45 column=cf:Country, timestamp=1732967733211, value=Asia
45 column=cf:Gas_Type, timestamp=1732967733211, value=N2O
45 column=cf:Industry, timestamp=1732967733211, value=EnergyCS
45 column=cf:Y_2022, timestamp=1732967733211, value=86
45 column=cf:Y_2023, timestamp=1732967733211, value=86
4 row(s) in 0.2000 seconds
hbase(main):006:0> █
```

- c. The command was used to view the largest amount of gas emitted, Carbon Dioxide by the Electricity, Gas, Steam and Air Conditioning Supply (EnergyCS) industry in Asia. The records show that emissions increased from 2022 to 2023.

```
hbase(main):007:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Country', '=', 'binary:Asia')
) AND (SingleColumnValueFilter('cf', 'Industry', '=', 'binary:EnergyCS')) AND (SingleColumnValueFilter('cf', 'Gas_Type',
=, 'binary:C02'))"}
ROW COLUMN+CELL
42 column=cf:Country, timestamp=1732967733211, value=Asia
42 column=cf:Gas_Type, timestamp=1732967733211, value=C02
42 column=cf:Industry, timestamp=1732967733211, value=EnergyCS
42 column=cf:Y_2022, timestamp=1732967733211, value=9769
42 column=cf:Y_2023, timestamp=1732967733211, value=9785
1 row(s) in 0.0440 seconds
hbase(main):008:0> █
```

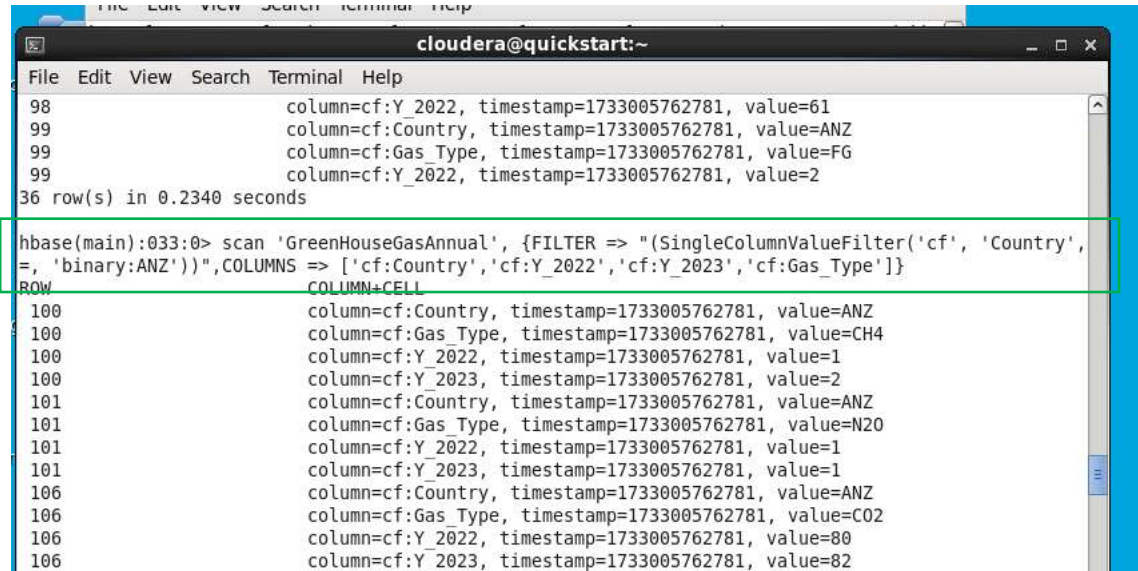
- d. The command was used to view the Carbon Dioxide emissions for 2022 across all countries, displaying only the country, gas type, and the emissions for 2022 (y\_2022).

```
hbase(main):012:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Gas_Type', '=', 'binary:C02'))", COLUMNS => ['cf:Country', 'cf:Y_2022', 'cf:Gas_Type']}
ROW COLUMN+CELL
1 column=cf:Country, timestamp=1732967733211, value=Africa
1 column=cf:Gas_Type, timestamp=1732967733211, value=C02
1 column=cf:Y_2022, timestamp=1732967733211, value=19
10 column=cf:Country, timestamp=1732967733211, value=Africa
10 column=cf:Gas_Type, timestamp=1732967733211, value=C02
10 column=cf:Y_2022, timestamp=1732967733211, value=345
106 column=cf:Country, timestamp=1732967733211, value=ANZ
106 column=cf:Gas_Type, timestamp=1732967733211, value=C02
106 column=cf:Y_2022, timestamp=1732967733211, value=80
110 column=cf:Country, timestamp=1732967733211, value=ANZ
110 column=cf:Gas_Type, timestamp=1732967733211, value=C02
110 column=cf:Y_2022, timestamp=1732967733211, value=1
114 column=cf:Country, timestamp=1732967733211, value=EA
114 column=cf:Gas_Type, timestamp=1732967733211, value=C02
114 column=cf:Y_2022, timestamp=1732967733211, value=94
117 column=cf:Country, timestamp=1732967733211, value=EA
117 column=cf:Gas_Type, timestamp=1732967733211, value=C02
117 column=cf:Y_2022, timestamp=1732967733211, value=285
120 column=cf:Country, timestamp=1732967733211, value=EA
120 column=cf:Gas_Type, timestamp=1732967733211, value=C02
120 column=cf:Y_2022, timestamp=1732967733211, value=6553
123 column=cf:Country, timestamp=1732967733211, value=EA
123 column=cf:Gas_Type, timestamp=1732967733211, value=C02
123 column=cf:Y_2022, timestamp=1732967733211, value=4642
127 column=cf:Country, timestamp=1732967733211, value=EA
127 column=cf:Gas_Type, timestamp=1732967733211, value=C02
127 column=cf:Y_2022, timestamp=1732967733211, value=735
130 column=cf:Country, timestamp=1732967733211, value=EA
130 column=cf:Gas_Type, timestamp=1732967733211, value=C02
130 column=cf:Y_2022, timestamp=1732967733211, value=220
134 column=cf:Country, timestamp=1732967733211, value=EA
134 column=cf:Gas_Type, timestamp=1732967733211, value=C02
134 column=cf:Y_2022, timestamp=1732967733211, value=830
14 column=cf:Country, timestamp=1732967733211, value=Africa
```

- e. The command was used to view the records of all gases emitted in 2022, as it holds the highest emission levels compared to 2023.

```
hbase(main):016:0> scan 'GreenHouseGasAnnual', {COLUMN => 'cf:Y_2022'}
ROW                                COLUMN+CELL
1                                  column=cf:Y_2022, timestamp=1732967733211, value=19
10                                 column=cf:Y_2022, timestamp=1732967733211, value=345
100                                column=cf:Y_2022, timestamp=1732967733211, value=1
101                                column=cf:Y_2022, timestamp=1732967733211, value=1
106                                column=cf:Y_2022, timestamp=1732967733211, value=80
107                                column=cf:Y_2022, timestamp=1732967733211, value=2
108                                column=cf:Y_2022, timestamp=1732967733211, value=1
109                                column=cf:Y_2022, timestamp=1732967733211, value=1
11                                 column=cf:Y_2022, timestamp=1732967733211, value=80
110                                column=cf:Y_2022, timestamp=1732967733211, value=1
111                                column=cf:Y_2022, timestamp=1732967733211, value=1
112                                column=cf:Y_2022, timestamp=1732967733211, value=14
113                                column=cf:Y_2022, timestamp=1732967733211, value=1
114                                column=cf:Y_2022, timestamp=1732967733211, value=94
115                                column=cf:Y_2022, timestamp=1732967733211, value=691
116                                column=cf:Y_2022, timestamp=1732967733211, value=266
117                                column=cf:Y_2022, timestamp=1732967733211, value=285
118                                column=cf:Y_2022, timestamp=1732967733211, value=1
119                                column=cf:Y_2022, timestamp=1732967733211, value=1
12                                 column=cf:Y_2022, timestamp=1732967733211, value=4
120                                column=cf:Y_2022, timestamp=1732967733211, value=6553
121                                column=cf:Y_2022, timestamp=1732967733211, value=11
122                                column=cf:Y_2022, timestamp=1732967733211, value=53
123                                column=cf:Y_2022, timestamp=1732967733211, value=4642
124                                column=cf:Y_2022, timestamp=1732967733211, value=387
125                                column=cf:Y_2022, timestamp=1732967733211, value=15
126                                column=cf:Y_2022, timestamp=1732967733211, value=31
127                                column=cf:Y_2022, timestamp=1732967733211, value=735
128                                column=cf:Y_2022, timestamp=1732967733211, value=951
129                                column=cf:Y_2022, timestamp=1732967733211, value=1
13                                 column=cf:Y_2022, timestamp=1732967733211, value=19
```

- f. Australia and New Zealand emitted the least gas among the 6 regions in both 2022 and 2023 years.



```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
98      column=cf:Y_2022, timestamp=1733005762781, value=61  
99      column=cf:Country, timestamp=1733005762781, value=ANZ  
99      column=cf:Gas_Type, timestamp=1733005762781, value=FG  
99      column=cf:Y_2022, timestamp=1733005762781, value=2  
36 row(s) in 0.2340 seconds  
hbase(main):033:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Country',  
=, 'binary:ANZ'))", COLUMNS => ['cf:Country', 'cf:Y_2022', 'cf:Y_2023', 'cf:Gas_Type']}  
ROW      COLUMN+CELL  
100      column=cf:Country, timestamp=1733005762781, value=ANZ  
100      column=cf:Gas_Type, timestamp=1733005762781, value=CH4  
100      column=cf:Y_2022, timestamp=1733005762781, value=1  
100      column=cf:Y_2023, timestamp=1733005762781, value=2  
101      column=cf:Country, timestamp=1733005762781, value=ANZ  
101      column=cf:Gas_Type, timestamp=1733005762781, value=N2O  
101      column=cf:Y_2022, timestamp=1733005762781, value=1  
101      column=cf:Y_2023, timestamp=1733005762781, value=1  
106      column=cf:Country, timestamp=1733005762781, value=ANZ  
106      column=cf:Gas_Type, timestamp=1733005762781, value=CO2  
106      column=cf:Y_2022, timestamp=1733005762781, value=80  
106      column=cf:Y_2023, timestamp=1733005762781, value=82
```



g. The largest amount of GreenHouse Gas emitted by Australia and New Zealand is Carbon Dioxide.

```
hbase(main):002:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Country',  
=, 'binary:ANZ')) AND (SingleColumnValueFilter('cf', 'Gas_Type',=, 'binary:CO2'))",COLUMNS => ['cf:C  
ountry','cf:Y_2022','cf:Y_2023','cf:Gas_Type']}]
```

| ROW | COLUMN+CELL                                            |
|-----|--------------------------------------------------------|
| 106 | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 106 | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 106 | column=cf:Y_2022, timestamp=1733005762781, value=80    |
| 106 | column=cf:Y_2023, timestamp=1733005762781, value=82    |
| 110 | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 110 | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 110 | column=cf:Y_2022, timestamp=1733005762781, value=1     |
| 110 | column=cf:Y_2023, timestamp=1733005762781, value=1     |
| 74  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 74  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 74  | column=cf:Y_2022, timestamp=1733005762781, value=12    |
| 74  | column=cf:Y_2023, timestamp=1733005762781, value=12    |
| 78  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 78  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 78  | column=cf:Y_2022, timestamp=1733005762781, value=4     |
| 78  | column=cf:Y_2023, timestamp=1733005762781, value=4     |
| 82  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 82  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 82  | column=cf:Y_2022, timestamp=1733005762781, value=171   |
| 82  | column=cf:Y_2023, timestamp=1733005762781, value=174   |
| 86  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 86  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 86  | column=cf:Y_2022, timestamp=1733005762781, value=61    |
| 86  | column=cf:Y_2023, timestamp=1733005762781, value=63    |
| 90  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 90  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 90  | column=cf:Y_2022, timestamp=1733005762781, value=47    |
| 90  | column=cf:Y_2023, timestamp=1733005762781, value=46    |
| 94  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 94  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 94  | column=cf:Y_2022, timestamp=1733005762781, value=5     |
| 94  | column=cf:Y_2023, timestamp=1733005762781, value=6     |
| 98  | column=cf:Country, timestamp=1733005762781, value=ANZ  |
| 98  | column=cf:Gas_Type, timestamp=1733005762781, value=CO2 |
| 98  | column=cf:Y_2022, timestamp=1733005762781, value=61    |
| 98  | column=cf:Y_2023, timestamp=1733005762781, value=62    |

9 row(s) in 0.0920 seconds

h. Fluorinated gases, FG is the least emitted gas across all region for both years.

```
hbase(main):001:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Gas_Type', '=', 'binary:FG'))", COLUMNS => ['cf:Country', 'cf:Gas_Type', 'cf:Y_2022', 'cf:Y_2023']}
ROW COLUMN+CELL
107 column=cf:Country, timestamp=1733005762781, value=ANZ
107 column=cf:Gas_Type, timestamp=1733005762781, value=FG
107 column=cf:Y_2022, timestamp=1733005762781, value=2
107 column=cf:Y_2023, timestamp=1733005762781, value=2
11 column=cf:Country, timestamp=1733005762781, value=Africa
11 column=cf:Gas_Type, timestamp=1733005762781, value=FG
11 column=cf:Y_2022, timestamp=1733005762781, value=80
11 column=cf:Y_2023, timestamp=1733005762781, value=85
111 column=cf:Country, timestamp=1733005762781, value=ANZ
111 column=cf:Gas_Type, timestamp=1733005762781, value=FG
111 column=cf:Y_2022, timestamp=1733005762781, value=1
111 column=cf:Y_2023, timestamp=1733005762781, value=1
124 column=cf:Country, timestamp=1733005762781, value=EA
124 column=cf:Gas_Type, timestamp=1733005762781, value=FG
124 column=cf:Y_2022, timestamp=1733005762781, value=387
124 column=cf:Y_2023, timestamp=1733005762781, value=399
131 column=cf:Country, timestamp=1733005762781, value=EA
131 column=cf:Gas_Type, timestamp=1733005762781, value=FG
131 column=cf:Y_2022, timestamp=1733005762781, value=23
131 column=cf:Y_2023, timestamp=1733005762781, value=24
135 column=cf:Country, timestamp=1733005762781, value=EA
135 column=cf:Gas_Type, timestamp=1733005762781, value=FG
135 column=cf:Y_2022, timestamp=1733005762781, value=23
135 column=cf:Y_2023, timestamp=1733005762781, value=24
143 column=cf:Country, timestamp=1733005762781, value=EA
143 column=cf:Gas_Type, timestamp=1733005762781, value=FG
143 column=cf:Y_2022, timestamp=1733005762781, value=44
143 column=cf:Y_2023, timestamp=1733005762781, value=44
150 column=cf:Country, timestamp=1733005762781, value=EU
150 column=cf:Gas_Type, timestamp=1733005762781, value=FG
150 column=cf:Y_2022, timestamp=1733005762781, value=1
150 column=cf:Y_2023, timestamp=1733005762781, value=1
154 column=cf:Country, timestamp=1733005762781, value=EU
154 column=cf:Gas_Type, timestamp=1733005762781, value=FG
...
```



- i. The following screenshot all the gases emitted by mining industry.

```
hbase(main):001:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Industry',=, 'binary:Minin
g'))", COLUMN => ['cf:Industry', 'cf:Gas_Type', 'cf:Y_2022', 'cf:Y_2023']}
ROW COLUMN+CELL
127 column=cf:Gas_Type, timestamp=1733005762781, value=C02
127 column=cf:Industry, timestamp=1733005762781, value=Mining
127 column=cf:Y_2022, timestamp=1733005762781, value=735
127 column=cf:Y_2023, timestamp=1733005762781, value=743
128 column=cf:Gas_Type, timestamp=1733005762781, value=CH4
128 column=cf:Industry, timestamp=1733005762781, value=Mining
128 column=cf:Y_2022, timestamp=1733005762781, value=951
128 column=cf:Y_2023, timestamp=1733005762781, value=1010
129 column=cf:Gas_Type, timestamp=1733005762781, value=N2O
129 column=cf:Industry, timestamp=1733005762781, value=Mining
129 column=cf:Y_2022, timestamp=1733005762781, value=1
129 column=cf:Y_2023, timestamp=1733005762781, value=1
14 column=cf:Gas_Type, timestamp=1733005762781, value=C02
14 column=cf:Industry, timestamp=1733005762781, value=Mining
14 column=cf:Y_2022, timestamp=1733005762781, value=99
14 column=cf:Y_2023, timestamp=1733005762781, value=98
15 column=cf:Gas_Type, timestamp=1733005762781, value=CH4
15 column=cf:Industry, timestamp=1733005762781, value=Mining
15 column=cf:Y_2022, timestamp=1733005762781, value=440
15 column=cf:Y_2023, timestamp=1733005762781, value=470
16 column=cf:Gas_Type, timestamp=1733005762781, value=N2O
16 column=cf:Industry, timestamp=1733005762781, value=Mining
16 column=cf:Y_2022, timestamp=1733005762781, value=0
16 column=cf:Y_2023, timestamp=1733005762781, value=0
165 column=cf:Gas_Type, timestamp=1733005762781, value=C02
165 column=cf:Industry, timestamp=1733005762781, value=Mining
165 column=cf:Y_2022, timestamp=1733005762781, value=183
165 column=cf:Y_2023, timestamp=1733005762781, value=185
166 column=cf:Gas_Type, timestamp=1733005762781, value=FG
166 column=cf:Industry, timestamp=1733005762781, value=Mining
166 column=cf:Y_2022, timestamp=1733005762781, value=0
166 column=cf:Y_2023, timestamp=1733005762781, value=0
167 column=cf:Gas_Type, timestamp=1733005762781, value=CH4
167 column=cf:Industry, timestamp=1733005762781, value=Mining
167 column=cf:Y_2022, timestamp=1733005762781, value=199
167 column=cf:Y_2023, timestamp=1733005762781, value=228
168 column=cf:Gas_Type, timestamp=1733005762781, value=N2O
168 column=cf:Industry, timestamp=1733005762781, value=Mining
168 column=cf:Y_2022, timestamp=1733005762781, value=1
168 column=cf:Y_2023, timestamp=1733005762781, value=1
202 column=cf:Gas_Type, timestamp=1733005762781, value=C02
202 column=cf:Industry, timestamp=1733005762781, value=Mining
202 column=cf:Y_2022, timestamp=1733005762781, value=202
202 column=cf:Y_2023, timestamp=1733005762781, value=202
```

- j. The largest emitted gas is Carbon Dioxide and the industry contributed least for Carbon Dioxide is Water supply; sewerage, waste management and remediation activities.

```
hbase(main):002:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Gas_Type',  
,=, 'binary:C02')) AND (SingleColumnValueFilter('cf', 'Industry',=, 'binary:WWS'))", COLUMN => ['cf:I  
ndustry', 'cf:Gas_Type', 'cf:Y_2022', 'cf:Y_2023']}
```

| ROW | COLUMN+CELL                                            |
|-----|--------------------------------------------------------|
| 110 | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 110 | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 110 | column=cf:Y_2022, timestamp=1733005762781, value=1     |
| 110 | column=cf:Y_2023, timestamp=1733005762781, value=1     |
| 146 | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 146 | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 146 | column=cf:Y_2022, timestamp=1733005762781, value=44    |
| 146 | column=cf:Y_2023, timestamp=1733005762781, value=45    |
| 185 | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 185 | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 185 | column=cf:Y_2022, timestamp=1733005762781, value=41    |
| 185 | column=cf:Y_2023, timestamp=1733005762781, value=41    |
| 221 | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 221 | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 221 | column=cf:Y_2022, timestamp=1733005762781, value=27    |
| 221 | column=cf:Y_2023, timestamp=1733005762781, value=27    |
| 31  | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 31  | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 31  | column=cf:Y_2022, timestamp=1733005762781, value=8     |
| 31  | column=cf:Y_2023, timestamp=1733005762781, value=7     |
| 70  | column=cf:Gas_Type, timestamp=1733005762781, value=C02 |
| 70  | column=cf:Industry, timestamp=1733005762781, value=WWS |
| 70  | column=cf:Y_2022, timestamp=1733005762781, value=86    |
| 70  | column=cf:Y_2023, timestamp=1733005762781, value=91    |

6 row(s) in 0.1770 seconds

- k. Fluorinated gases, FG is the least emitted gas and the huge contributed is the Manufacturing industry.

```
hbase(main):004:0> scan 'GreenHouseGasAnnual', {FILTER => "(SingleColumnValueFilter('cf', 'Gas_Type'
,=, 'binary:FG')) AND (SingleColumnValueFilter('cf', 'Industry',=, 'binary:Manuf'))", COLUMN => ['cf:
Industry', 'cf:Gas_Type', 'cf:Y_2022', 'cf:Y_2023']}
ROW          COLUMN+CELL
11           column=cf:Gas_Type, timestamp=1733005762781, value=FG
11           column=cf:Industry, timestamp=1733005762781, value=Manuf
11           column=cf:Y_2022, timestamp=1733005762781, value=80
11           column=cf:Y_2023, timestamp=1733005762781, value=85
124          column=cf:Gas_Type, timestamp=1733005762781, value=FG
124          column=cf:Industry, timestamp=1733005762781, value=Manuf
124          column=cf:Y_2022, timestamp=1733005762781, value=387
124          column=cf:Y_2023, timestamp=1733005762781, value=399
162          column=cf:Gas_Type, timestamp=1733005762781, value=FG
162          column=cf:Industry, timestamp=1733005762781, value=Manuf
162          column=cf:Y_2022, timestamp=1733005762781, value=53
162          column=cf:Y_2023, timestamp=1733005762781, value=54
199          column=cf:Gas_Type, timestamp=1733005762781, value=FG
199          column=cf:Industry, timestamp=1733005762781, value=Manuf
199          column=cf:Y_2022, timestamp=1733005762781, value=4
199          column=cf:Y_2023, timestamp=1733005762781, value=4
47           column=cf:Gas_Type, timestamp=1733005762781, value=FG
47           column=cf:Industry, timestamp=1733005762781, value=Manuf
47           column=cf:Y_2022, timestamp=1733005762781, value=684
47           column=cf:Y_2023, timestamp=1733005762781, value=699
87           column=cf:Gas_Type, timestamp=1733005762781, value=FG
87           column=cf:Industry, timestamp=1733005762781, value=Manuf
87           column=cf:Y_2022, timestamp=1733005762781, value=1
87           column=cf:Y_2023, timestamp=1733005762781, value=1
6 row(s) in 0.1070 seconds

hbase(main):005:0> █
```

- l. The command was used to view the records starting from row 3, limiting the results to 10 records.

```
hbase(main):021:0> scan 'GreenHouseGasAnnual', {COLUMN => ['cf:Country', 'cf:Y_2023'], LIMIT => 10, STARTROW => '3'}
COLUMN=CELL
3      column=cf:Country, timestamp=1732967733211, value=Africa
3      column=cf:Y_2023, timestamp=1732967733211, value=265
30     column=cf:Country, timestamp=1732967733211, value=Africa
30     column=cf:Y_2023, timestamp=1732967733211, value=11
31     column=cf:Country, timestamp=1732967733211, value=Africa
31     column=cf:Y_2023, timestamp=1732967733211, value=7
32     column=cf:Country, timestamp=1732967733211, value=Africa
32     column=cf:Y_2023, timestamp=1732967733211, value=326
33     column=cf:Country, timestamp=1732967733211, value=Africa
33     column=cf:Y_2023, timestamp=1732967733211, value=16
34     column=cf:Country, timestamp=1732967733211, value=Asia
34     column=cf:Y_2023, timestamp=1732967733211, value=238
35     column=cf:Country, timestamp=1732967733211, value=Asia
35     column=cf:Y_2023, timestamp=1732967733211, value=8
36     column=cf:Country, timestamp=1732967733211, value=Asia
36     column=cf:Y_2023, timestamp=1732967733211, value=2155
37     column=cf:Country, timestamp=1732967733211, value=Asia
37     column=cf:Y_2023, timestamp=1732967733211, value=831
38     column=cf:Country, timestamp=1732967733211, value=Asia
38     column=cf:Y_2023, timestamp=1732967733211, value=439
10 row(s) in 0.0690 seconds
```

- m. The command was used to view the records in reverse order.

```
200 row(s) in 2.6120 seconds
hbase(main):002:0> scan 'GreenHouseGasAnnual', {REVERSED => true}
Icloudera@quickstart:~1 cloudera@quickstart:~ [Hue - HBase Browser ...
```

The bottom part of the output

```
100     column=cf:Gas_Type, timestamp=1732967733211, value=CH4
100     column=cf:Industry, timestamp=1732967733211, value=Households
100     column=cf:Y_2022, timestamp=1732967733211, value=1
100     column=cf:Y_2023, timestamp=1732967733211, value=2
10     column=cf:Country, timestamp=1732967733211, value=Africa
10     column=cf:Gas_Type, timestamp=1732967733211, value=CO2
10     column=cf:Industry, timestamp=1732967733211, value=Manuf
10     column=cf:Y_2022, timestamp=1732967733211, value=345
10     column=cf:Y_2023, timestamp=1732967733211, value=330
1     column=cf:Country, timestamp=1732967733211, value=Africa
1     column=cf:Gas_Type, timestamp=1732967733211, value=CO2
1     column=cf:Industry, timestamp=1732967733211, value=Agriculture
1     column=cf:Y_2022, timestamp=1732967733211, value=19
1     column=cf:Y_2023, timestamp=1732967733211, value=18
200 row(s) in 1.7840 seconds
hbase(main):003:0>
```

## 6. DELETEALL (DML)

Upon reviewing the output of Scan a., it was observed that the value for Ind\_HH was significantly larger compared to the others. After performing analysis in Excel, identified that this value represented the summation of all industries and households. As a result, all rows with "Ind\_HH" were deleted. A total of 24 records related to Industry and Households were removed.

```
hbase(main):063:0> deleteall 'GreenHouseGasAnnual', '102'
0 row(s) in 0.0090 seconds

hbase(main):064:0> deleteall 'GreenHouseGasAnnual', '102'
0 row(s) in 0.0120 seconds

hbase(main):065:0> deleteall 'GreenHouseGasAnnual', '103'
0 row(s) in 0.0060 seconds

hbase(main):066:0> deleteall 'GreenHouseGasAnnual', '104'
0 row(s) in 0.0070 seconds

hbase(main):067:0> deleteall 'GreenHouseGasAnnual', '105'
0 row(s) in 0.0060 seconds

hbase(main):068:0> deleteall 'GreenHouseGasAnnual', '105'
0 row(s) in 0.0040 seconds

hbase(main):069:0> deleteall 'GreenHouseGasAnnual', '138'
0 row(s) in 0.0070 seconds

hbase(main):070:0> deleteall 'GreenHouseGasAnnual', '139'
0 row(s) in 0.0130 seconds

hbase(main):071:0> deleteall 'GreenHouseGasAnnual', '140'
0 row(s) in 0.0060 seconds

hbase(main):072:0> deleteall 'GreenHouseGasAnnual', '141'
0 row(s) in 0.0030 seconds

hbase(main):073:0> deleteall 'GreenHouseGasAnnual', '177'
0 row(s) in 0.0020 seconds

hbase(main):074:0> deleteall 'GreenHouseGasAnnual', '178'
0 row(s) in 0.0140 seconds

hbase(main):075:0> deleteall 'GreenHouseGasAnnual', '179'
0 row(s) in 0.0070 seconds

hbase(main):076:0> deleteall 'GreenHouseGasAnnual', '180'
0 row(s) in 0.0020 seconds

hbase(main):077:0> deleteall 'GreenHouseGasAnnual', '213'
0 row(s) in 0.0080 seconds

hbase(main):078:0> deleteall 'GreenHouseGasAnnual', '214'
0 row(s) in 0.0030 seconds

hbase(main):079:0> deleteall 'GreenHouseGasAnnual', '215'
```



The number of rows after deleting the 24 records is just 200 records.

```
95 column=cf:Industry, timestamp=1732967733211, value=Others
96 column=cf:Y_2022, timestamp=1732967733211, value=0
96 column=cf:Y_2023, timestamp=1732967733211, value=0
97 column=cf:Country, timestamp=1732967733211, value=ANZ
97 column=cf:Gas_Type, timestamp=1732967733211, value=N2O
97 column=cf:Industry, timestamp=1732967733211, value=Others
97 column=cf:Y_2022, timestamp=1732967733211, value=0
97 column=cf:Y_2023, timestamp=1732967733211, value=0
98 column=cf:Country, timestamp=1732967733211, value=ANZ
98 column=cf:Gas_Type, timestamp=1732967733211, value=CO2
98 column=cf:Industry, timestamp=1732967733211, value=Households
98 column=cf:Y_2022, timestamp=1732967733211, value=61
98 column=cf:Y_2023, timestamp=1732967733211, value=62
99 column=cf:Country, timestamp=1732967733211, value=ANZ
99 column=cf:Gas_Type, timestamp=1732967733211, value=FG
99 column=cf:Industry, timestamp=1732967733211, value=Households
99 column=cf:Y_2022, timestamp=1732967733211, value=2
99 column=cf:Y_2023, timestamp=1732967733211, value=2
200 row(s) in 0.7260 seconds
```

## 7. DISABLE (DDL)

This command disables the GreenHouseGasAnnual table, preventing any further modifications. Executed this command when I was away from my machine.

```
hbase(main):008:0> disable 'GreenHouseGasAnnual'
0 row(s) in 2.3210 seconds

hbase(main):009:0> get 'GreenHouseGasAnnual','1'
COLUMN                                CELL
ERROR: GreenHouseGasAnnual is disabled.
```

## 8. ENABLE (DDL)

This command enables the GreenHouseGasAnnual table, allowing modifications to be made.

```
hbase(main):010:0> enable 'GreenHouseGasAnnual'
0 row(s) in 1.3040 seconds

hbase(main):011:0> get 'GreenHouseGasAnnual','1'
COLUMN                                CELL
cf:Country                            timestamp=1733003742749, value=Africa
cf:Gas_Type                           timestamp=1733003742749, value=CO2
cf:Industry                           timestamp=1733003742749, value=Agriculture
cf:Y_2022                             timestamp=1733003742749, value=19
cf:Y_2023                             timestamp=1733003742749, value=18
5 row(s) in 0.0220 seconds

hbase(main):012:0> █
```

## 9. DROP (DDL)

This command deletes the GreenHouseGasAnnual table from the database.

```
hbase(main):013:0> disable 'GreenHouseGasAnnual'
0 row(s) in 2.3490 seconds

hbase(main):014:0> drop 'GreenHouseGasAnnual'
0 row(s) in 1.3910 seconds

hbase(main):015:0> list
TABLE
0 row(s) in 0.0280 seconds

=> []
hbase(main):016:0> █
```

## 10. EXIST (DDL)

Since we have deleted the table, the output shows that the table does not exist.

```
0 row(s) in 0.0330 seconds

=> []
hbase(main):016:0> exists 'GreenHouseGasAnnual'
Table GreenHouseGasAnnual does not exist
0 row(s) in 0.0330 seconds

hbase(main):017:0> █
```

END